1 Introduction

HydroFlex is a two-stage stochastic linear program / mixed integer linear program used for energy system optimization. These optimization models are used to identify the least cost combination of generation and storage assets capable of meeting the hourly electrical demand for a community, or to use these assets to respond to an energy market signal in order to maximize value. The model described her is based on the Distributed Energy Resource Optimization tool (DERopt) l[1] [1–4]. The transition to the HydroFlex name signifies a shift in focus towards hydroelectric generation assets where systems are designed to support a ”backbone” hydroelectric generator. In this new HydroFlex formulation, there is either an existing hydroelectric resource present where new generation and storage technologies are intended to improve the operational flexibility of the hydroelectric system, or new hydroelectric generation is known to be economically desirable and will be adopted in an optimization model. Hydroelectric systems included in this model include:

- Conventional dam
- Run-of-the-river
- Pumped hydro
- Hydrokinetic

The purpose of this document is to describe the formulation of the HydroFlex model from an optimization model building perspective. This will include both linear (LP) and mixed integer linear (MILP) versions of the model. The general model is formulated as a two-stage stochastic model, where the first stage captures technology adoption and sizing, and the second captures technology operation. This model can be converted to a deterministic model if a single operational scenario is considered. The time horizon in the model can be adjusted to fit the size of available data, but has a default hourly timestep with 8760 total hours, or an annual simulation.

2 Model Sets

Theses sets define the model structure:

- $t \in T$: Set of all timesteps in the simulation.
- $q \in Q$: Set of all timesteps t that fall within aTOU window fo determining demand charges ($Q \subseteq T$)
- $m \in M$: Set of all months in the simulation.

[1] https://github.com/rjflores2/DERopt

- n∈N: Set of all nodes with energy demand or energy infrastructure for transmitting or distributing energy through a network.
- s∈S: Set of all scenarios. Scenario s has an associated probability π_s of occuring where the sum of all π_s must equal 1.
- p∈P: Set of solar photovoltaic (PV) technologies, each with a different orientation or mounting system, or different solar PV technology.
- w∈W: Set of all wind powered technologies.
- r∈R: Set of all run-of-river (RoR) technologies.
- V = P∪W∪R: V represents the union of all renewable energy technologies that share similar constraint types
- k∈K: Set of all hydrokinetic technologies (HKT) and installation sites.
- f∈F: Set of all generators powered by conventional fuel captured with a linear capacity term.

$$\min Z = \Gamma^{Fixed} + \sum_{s \epsilon S} \pi_s \Gamma_s^{Ops} \tag{1}$$

$$\begin{aligned}\Gamma^{Fixed} = \sum_{n \epsilon N} \Bigg(& \sum_{p \epsilon P} \left(C_p^{CAP} + C_p^{FOM}\right) P_{p,n} + \sum_{w \epsilon W} \left(C_W^{CAP} + C_w^{FOM}\right) P_{w,n} \\ & + \sum_{r \epsilon R} \left(C_r^{CAP} + C_r^{FOM}\right) P_{r,n} + \sum_{k \epsilon K} \left(C_k^{CAP} + C_k^{FOM}\right) n_{k,n} \\ & + \sum_{f \epsilon F} \left(C_f^{CAP} + C_f^{FOM}\right) P_{f,n} + \sum_{g \epsilon G} \left(C_g^{CAP} + C_g^{FOM}\right) n_{g,n} \\ & + \left(C^{Batt\ CAP} + C^{Batt\ FOM}\right) E_n^{Batt} \\ & + \left(C^{Flow\ Batt\ CAP} + C^{Flow\ Batt\ FOM}\right) E_n^{Flow\ Batt} \\ & + \left(C^{Flow\ Sheet\ CAP} + C^{Flow\ Sheet\ FOM}\right) P_n^{Flow\ Batt} \\ & + \left(C^{Electrolyzer\ CAP} + C^{Electrolyzer\ \ FOM}\right) P_n^{Electrolyzer} \\ & + \left(C^{Fuel\ Cell\ CAP} + C^{Fuel\ Cell\ \ FOM}\right) P_n^{Fuel\ Cell} \\ & + \left(C^{H_2\ Tanks\ CAP} + C^{H_2\ Tanks\ \ FOM}\right) E_n^{H_2\ Tanks} \Bigg)\end{aligned} \tag{2}$$

$$\begin{aligned}\Gamma_s^{Ops} = \sum_{t \epsilon T} \sum_{n \epsilon N} \Bigg(& c_t^{Import} e_{n,s,t}^{Utility\ Import} + c_t^{Export} e_{n,s,t}^{Utility\ Export} + \sum_{k \epsilon K} c_k^{VOM} e_{n,k,s,t} \\ & + \sum_{f \epsilon F} \left(c_f^{VOM} + \frac{c_{f,n}^{fuel}}{\eta_f}\right) e_{n,f,s,t} + \sum_{g \epsilon G} \left(c_g^{VOM} + \frac{c_{g,n}^{fuel}}{\eta_g}\right) e_{n,g,s,t} + c_g^{VOM} \\ & + c_{Batt}^{VOM} \left(e_{n,s,t}^{Batt\ Charge} + e_{n,s,t}^{Batt\ Discharge}\right) \\ & + c_{Flow\ Batt}^{VOM} \left(e_{n,s,t}^{Flow\ Batt\ Charge} + e_{n,s,t}^{Flow\ Batt\ Discharge}\right) \\ & + C_{Electrolyzer}^{VOM} e_{n,s,t}^{Electrolyzer} + C_{Fuel\ Cell}^{VOM} e_{n,s,t}^{Fuel\ Cell} \Bigg) \\ & + \sum_{m \epsilon M} \sum_{n \epsilon N} \left(C_{n,s,m}^{DC\ Flat} P_{n,s,m}^{Monthly\ Max} + C_{n,s,m}^{DC\ TOU} P_{n,s,m}^{TOU\ Max}\right)\end{aligned} \tag{3}$$

- g∈G: Set of all generators powered by conventional fuel captured with a mixed integer linear capacity term.
- d∈D: Set of controllable hydroelectric units (conventional dam and pumped hydroelectric)

3 Objective Function

The goal of the general objective function is to minimize cost. Equation (1) breaks costs between the two stages in the stochastic optimization. Fixed costs, or first stage costs, are captured in Equation (2) and include equipment capital and fixed O&M costs (FOM). Second stage operational costs are captured in Equation (3), which captures all variable equipment costs, fuel costs, and utility electricity import, export, and demand charges.

4 Energy Balance Constraints

The cost function in Equation (1) is subject to energy balance, technology, and utility constraints. The first constraints are energy balances at each node through time, ensuring that electricity generation, utility imports, and storage discharge equal loads, utility exports, and storage charging (Equation (4)). A similar constraint is applied to hydrogen systems, ensuring that electrolyzer production and hydrogen tank discharge equals hydrogen needed to operate the fuel cell and to charge the hydrogen storage tank.

$$e_{n,s,t}^{Utility\ Import} + \sum_{v \epsilon V} e_{n,v,s,t} + \sum_{k \epsilon K} e_{n,k,s,t} + \sum_{f \epsilon F} e_{n,f,s,t} + \sum_{g \epsilon G} e_{n,g,s,t} + \sum_{d \epsilon D} e_{n,D,s,t} + e_{n,s,t}^{Batt\ Discharge} + e_{n,s,t}^{Flow\ Batt\ Discharge} + e_{n,s,t}^{Fuel\ Cell} = Load_{n,s,t} + e_{n,s,t}^{Batt\ Charge} + e_{n,s,t}^{Flow\ Batt\ Charge} + e_{n,s,t}^{Electrolyzer} + e_{n,s,t}^{H_2\ Compression} + e_{n,s,t}^{Utility\ Export} \quad (4)$$

$$\eta_{Electrolyzer} e_{n,s,t}^{Electrolyzer} + e_{n,s,t}^{H_2 Tank\ Discharge} = \frac{e_{n,s,t}^{Fuel\ Cell}}{\eta_{Fuel\ Cell}} + e_{n,s,t}^{H_2 Tank\ Charge} \quad (5)$$

5 Utility Constraints

Utility constraints govern energy import and export, and demand charge behavior. These constraints are highly specific to individual utility rates and tariffs.

5.1 Demand Charges

Flat and time of use demand charges are governed by Equations (6) and (7) respectively. In each constraint, the demand charge for month m is determined based off of the maximum

utility import across all time t that belongs in month m. For time of use demand charges, the subset t in month m is reduced to the subset q, or all hours that occur during the time of use window for which the TOU demand charge applies.

$$e_{n,s,t}^{Utility\ Import} \leq P_{n,s,m}^{Monthly\ Max}, t \subseteq m \tag{6}$$

$$e_{n,s,q}^{Utility\ Import} \leq P_{n,s,m}^{TOU\ Max}, q \subseteq t \subseteq m \tag{7}$$

5.2 Southern California Edison

The current HydroFlex model includes constraints specific to the Southern California Edison electrical utility. These constraints primarily govern electric utility exports. Equation (8) requires that net utility imports exceed exports, and Equation (9) requires that the net energy charge portion of a utility customers electric bill never drops below zero.

$$0 \leq \sum_{t \epsilon T} e_{n,s,t}^{Utility\ Import} + e_{n,s,t}^{Utility\ Export} \tag{8}$$

$$0 \leq \sum_{t \epsilon T} c_t^{Import} e_{n,s,t}^{Utility\ Import} + c_t^{Export} e_{n,s,t}^{Utility\ Export} \tag{9}$$

6 Solar, Wind, and Run-of-the-River

Solar, wind, and run-of-the-river technologies share a similar technology constraint structure. These technologies are governed by two constraints. Equation **Error! Reference source not found.** limits production from these technologies to the installed capacity multiplied by the resource availability at node n at time t. This constraint also includes existing, or legacy generation of type v. Equation limits new capacity based on Equation (11). This value can be based on any technical limitation but is typically set based on maximum area (e.g., roof top, open field, stream width). This model also covers situations where a site has existing hydrokinetic generation that cannot be expanded (e.g., $\bar{P}_{v,n} = 0$).

$$e_{n,v,s,t} \leq \rho_{n,v,s,t}\left(P_{v,n} + P_{v,n}^{Legacy}\right) \tag{10}$$

$$P_{v,n} \leq \bar{P}_{v,n} \tag{11}$$

7 Hydrokinetic Generation

Hydrokinetic generation differs from conventional run-of-river hydropower in that hydrokinetic systems extract energy directly from flowing-water currents, typically without creating hydraulic head or diverting flow through a penstock. Hydrokinetic units are often smaller and modular, with factory-manufactured turbine-generator equipment, whereas run-of-river facilities are commonly more site-specific installations that use a diversion and hydraulic head and may employ larger, custom-engineered equipment. Due to this, hydrokinetic generation is treated differently than run-of-the-river hydroelectric generation where we assume the hydrokinetic generation is installed in generator increments (i.e., the decision to adopt a hydrokinetic generator is an integer decision).

Much like solar, wind, and run-of-river, hydrokinetic generation is limited by the installed capacity and resource potential in Equation (12). This equation, however, bases installed capacity off of number of installed hydrokinetic generators $n_{k,n}$ multiplied by the capacity of each generator P_{HKT}. Number of hydrokinetic generators are limited in Equation (13).

$$e_{n,k,s,t} \leq \rho_{n,k,s,t} P_{HKT} n_{k,n} \tag{12}$$

$$n_{k,n} \leq \bar{n}_{k,n} \tag{13}$$

8 Generators, Electrolyzers, and Fuel Cells

Fuel-fired generators, electrolyzers, and fuel cells share a common structure. Each converts one energy carrier into another at a fixed efficiency, each is limited by installed capacity, and each may be subject to a minimum loading requirement when operating. This section defines three formulations that apply to all of them. A case assigns one formulation to each technology.

Let $j \in J$ index these technologies, where J contains each fuel-fired generator, the electrolyzer, and the fuel cell. Let $e_{n,j,s,t}$ be the flow that installed capacity constrains, in kWh per timestep. The meaning of $e_{n,j,s,t}$ and its coupling to the energy balances differ by technology and are given in Section

8.1 Linear Formulation

Capacity is continuous and operation is unrestricted below capacity. Equation (14) limits the flow to installed capacity, including legacy capacity. Equation (15) limits new adoption.

$$e_{n,j,s,t} \leq \left(P_{j,n} + P_j^{Legacy}, n\right) \tag{14}$$

$$P_{j,n} \leq \bar{P}_{j,n} \tag{15}$$

This formulation permits operation at any load down to zero. It is appropriate where part-load efficiency penalties and minimum stable output are not material, and it is the only formulation that preserves linearity, so it is the formulation used when dual prices are required. This model is useful for evaluating if a technology is economically viable for a specific application since additional constraints that realistically restrict generator operation reduce equipment value.

8.2 Linear Formulation with Binary Operating State

Capacity remains continuous, and a binary variable $y_{n,j,s,t}$ records whether the technology is operating in each timestep ($y_{n,j,s,t} \in \{0,1\}$). Equations (14) and (15) still apply. Equation (16) forces the flow to zero when the technology is off. Equation (17) enforces the minimum loading fraction β_j when it is on.

$$e_{n,j,s,t} \leq M_{n,j}\, y_{n,j,s,t} \tag{16}$$

$$e_{n,j,s,t} \geq \beta_j\left(P_{j,n} + P_{j,n}^{\text{Legacy}}\right) - M_{n,j}\left(1 - y_{n,j,s,t}\right) \tag{17}$$

$M_{n,j}$ is a scalar large enough that Equation (16) does not limit output when the technology is on. It is set from the peak load at the node, shown in Equation (18).

$$M_{n,j} = \Lambda_j \max_{s \in S,\, t \in T}\left(Load_{n,s,t}\right) \tag{18}$$

Λ_j is 1 for generators and for the fuel cell, reflecting an assumption that neither is sized above peak load. For the electrolyzer Λ_j is larger, because a large solar or hydrokinetic installation may support an electrolyzer well above peak electrical demand.

This formulation is useful for when the target load is relatively small such that a single generator would be adopted.

8.3 Mixed Integer Formulation

Capacity is adopted in whole units and the number of units operating in each timestep is an integer decision. Each unit has capacity P_j^{Unit}. Equation (19) limits unit adoption.

Equation (20) limits operating units to installed units. Equations (21) and (22) bound the flow between the minimum and maximum output of the operating units.

$$u_{j,n} \leq \bar{u}_{j,n} \tag{19}$$

$$v_{n,j,s,t} \leq u_{j,n} + u_{j,n}^{\text{Legacy}} \tag{20}$$

$$e_{n,j,s,t} \leq P_j^{\text{Unit}}\, v_{n,j,s,t} \tag{21}$$

$$e_{n,j,s,t} \geq \beta_j\, P_j^{\text{Unit}}\, v_{n,j,s,t} \tag{22}$$

This formulation represents a fleet of identical units and captures the operating cost of running several units at part load rather than fewer units at full load. It requires no large scalar, because the integer variable $v_{n,j,s,t}$ sets both bounds directly. The minimum loading constraint is therefore exact rather than relaxed, which is the main modeling advantage over

8.4 Technology Coupling

The three formulations are identical in form. The technologies differ in what $e_{n,j,s,t}$ represents and in how it enters the balances of the energy balance equations in Section 4. These differences are summarized in Table 1.

Table 1: Mapping between generators, electrolyzers, and fuel cells describe in this section and these technologies in the model

Technology	$e_{n,j,s,t}$ represents	Electricity balance	Hydrogen balance	Fuel consumed
Fuel-fired generator	Electricity generated	Source	Not applicable	$e_{n,j,s,t}/\eta_j$
Electrolyzer	Electricity consumed	Sink	Produced, $\eta_{\text{Electrolyzer}}\, e_{n,j,s,t}$	Not applicable
Fuel cell	Electricity generated	Source	Consumed, $e_{n,j,s,t}/\eta_{\text{Fuel Cell}}$	Not applicable

Capacity is stated in kW of the flow that $e_{n,j,s,t}$ represents. Generator and fuel cell capacity are in kW electric output. Electrolyzer capacity is in kW electric input, not in kW of hydrogen chemical energy. Fuel consumption in fuel-fired generators is not tracked by a balance constraint because fuel is assumed to be available on demand. As a result, fuel enters only the operating cost in Equation (3).

9 Energy Storage

Electrical batteries, flow batteries, and hydrogen tanks share a common formulation. Each system retains a state of charge that can be changed by either charging or discharging the energy storage system. Overtime, storage state of charge may naturally decrease due to self-discharge. The technologies differ in which energy carrier they store, in how charge and discharge power is sized, and in whether charging requires an additional input.

Let ℓ denote a storage technology, where $\ell \in \{Batt, Flow\ Batt, H_2\ Tank\}$. A case selects one electrical storage technology and may additionally include hydrogen storage. Let $E^{\mathrm{SOC},\ell}_{n,s,t}$ be the stored inventory, and let $e^{\ell\ Charge}_{n,s,t}$ and $e^{\ell\ Discharge}_{n,s,t}$ be the charge and discharge flows in kWh per timestep. Total installed energy capacity is captured in Equation **Error! Reference source not found.**.

$$\tilde{E}^{\ell}_{n} = E^{\ell}_{n} + E^{\mathrm{Legacy},\ell}_{n} \tag{23}$$

9.1 Common Constraints

Equation (24) is the inventory balance. Inventory carried from the prior timestep is reduced by the retention factor α_{ℓ}, increased by charging at efficiency η^{ℓ}_{Charge}, and decreased by discharging at efficiency $\eta^{\ell}_{Discharge}$. The index $t-1$ refers to a prior timestep, linking current energy storage state of charge to the prior time step. At the start of the time vector in the simulation, this prior time step is taken as T, the final time step in the simulation. As a result, initial state of charge is linked to final state of charge in the simulation.

$$E^{\mathrm{SOC},\ell}_{n,s,t} = \alpha_{\ell} E^{\mathrm{SOC},\ell}_{n,s,t-1} + \eta^{\ell}_{Charge}\, e^{\ell\ Charge}_{n,s,t} - \frac{e^{\ell\ Discharge}_{n,s,t}}{\eta^{\ell}_{Discharge}} \tag{24}$$

Equation (25) limits inventory to a usable window expressed as fractions of installed energy capacity.

$$\underline{\sigma}_{\ell}\ \tilde{E}_{n}^{\ell} \leq E_{n,s,t}^{\mathrm{SOC},\ell} \leq\ \bar{\sigma}_{\ell}\ \tilde{E}_{n}^{\ell} \tag{25}$$

The current model allows the user to specify initial storage state of charge. This constraint, however, is typically not active during energy storage sizing because a user will likely not know the financial impact of a predetermined, likely arbitrary, and potentially suboptimal initial state of charge constraint.

9.2 Charge and Discharge Constraints

Two variants exist. The choice reflects whether power and energy capacity are physically coupled in the device. For batteries and hydrogen tanks, charge and discharge power scale with installed energy capacity through a rate limit μ^{ℓ}, expressed as kW per kWh of capacity.

$$e_{n,s,t}^{\ell\ Charge} \leq \mu_{Charge}^{\ell}\ \tilde{E}_{n}^{\ell} \tag{26}$$

$$e_{n,s,t}^{\ell\ Discharge} \leq \mu_{Discharge}^{\ell}\ \tilde{E}_{n}^{\ell} \tag{27}$$

A flow battery stores energy in electrolyte tanks and converts it in a separate stack, so power and energy are sized independently. Equation (28) is analogous to Equation (23) where we related total flow battery capacity to newly adopted plus legacy flow battery power capacity. Equations (26) and (27) are replaced by Equations (29) and (30), and a power capacity term is added to Γ^{Fixed}.

$$\tilde{P}_{n}^{\ell} = P_{n}^{\ell} + P_{n}^{\mathrm{Legacy},\ell} \tag{28}$$

$$e_{n,s,t}^{\ell\ Charge} \leq \tilde{P}_{n}^{\ell} \tag{29}$$

$$e_{n,s,t}^{\ell\ Discharge} \leq \tilde{P}_{n}^{\ell} \tag{30}$$

9.3 Hydrogen Storage Compression

Compressing hydrogen into storage consumes electricity in proportion to the hydrogen charged. Equation (31) defines that demand, which enters the electricity balance as a sink.

$$e_{n,s,t}^{H_2\ Compression} = \kappa\ e_{n,s,t}^{H_2\ Charge} \tag{31}$$

Compression electricity carries no separate cost term. It is priced implicitly through whatever generation or import serves it. Discharging hydrogen from storage requires no electrical input.

9.4 Energy Storage Coupling

Table 2: Mapping between generators, electrolyzers, and fuel cells describe in this section and these technologies in the model

Technology	Inventory units	Charge flow	Discharge flow	Additional electricity demand
Battery	kWh electric	Electricity sink	Electricity source	None
Flow battery	kWh electric	Electricity sink	Electricity source	None
Hydrogen tank	kWh H_2, LHV	Hydrogen sink	Hydrogen source	$\kappa\, e_{n,s,t}^{H_2\ Charge}$

10Controllable Hydroelectric Generation

Conventional dam hydro and pumped storage hydro share one formulation. Both hold an inventory of stored energy potential, both release it through a turbine to generate electricity, and both receive natural inflow that arrives independently of any decision. They differ in one respect. Pumped storage can add to inventory by consuming electricity, while conventional dam hydro cannot. Setting pump capacity to zero reduces the formulation to a conventional dam, so no separate model is required.

Hydroelectric capacity is treated as existing. No hydro adoption decision is included, so all hydro capacity terms are parameters and hydro contributes nothing to Γ^{Fixed}.

Let $d \in D$index controllable hydro units, and let $D_n \subseteq D$be the units located at node n. Inventory is expressed in kWh of stored energy potential, consistent with the treatment of all other storage in Section 9

10.1 Inventory Balance

Inventory balance tracks water flow in and out of a reservoir. The current HydroFlex model uses energy in kWh as key decision variables for all technologies, and uses kWh to track energy stored in a reservoir. A dedicated hydroelectric model would use reservoir depth or volume of water stored. However, we use kWh in order to keep a common unit between technologies to enable quick conversion between inputs and outputs between different technologies.

Equation (32) is the inventory balance. Inventory carried from the prior timestep is reduced by the retention factor α_d, which represents evaporation and seepage. Inventory increases through natural inflow $\iota_{d,s,t}$ and through pumping at efficiency η^{D}_{Pump}. It decreases through turbine release and through spill.

$$E^{\mathrm{SOC},D}_{d,s,t} = \alpha_d E^{\mathrm{SOC},D}_{d,s,t-1} + \iota_{d,s,t} + \eta^{D}_{Pump}\, e^{D\ Pump}_{d,s,t} - \frac{e^{D\ Gen}_{d,s,t}}{\eta^{D}_{Gen}} - \varsigma_{d,s,t} \tag{32}$$

$\iota_{d,s,t}$ is a parameter, not a decision variable. Inflow arrives whether or not the reservoir has room for it. $\varsigma_{d,s,t} \geq 0$ is spill, which is water released without passing through the turbine. Spill produces no electricity and carries no cost. It exists so that inflow exceeding the combined capacity of storage and turbine has somewhere to go. Without it, a wet period renders the model infeasible rather than spilling.

Equation (33) limits inventory to the usable range of the reservoir, expressed as fractions of total energy capacity $\tilde{E}^{D}_{d}$.

$$\underline{\sigma}_d\ \tilde{E}^{D}_{d} \leq E^{\mathrm{SOC},D}_{d,s,t} \leq \bar{\sigma}_d\ \tilde{E}^{D}_{d} \tag{33}$$

10.2 Boundary Conditions

Unlike battery storage, reservoir storage is often seasonal, and the treatment of the horizon boundary is a modeling choice that materially affects results. One of three options is selected per case and must be stated with any reported result.

- **Cyclic.** The index $t-1$ in Equation (32) refers to the final timestep when t is the first. Inventory returns to its starting level over the horizon, so the reservoir cannot be drawn down on net. This is appropriate for a long-run planning study.
- **Fixed endpoints.** Equation (32) applies for $t \geq 2$ only, and Equations (34) and (35) fix the starting and ending inventory.

$$E^{\mathrm{SOC},D}_{d,s,1} = \gamma^{\mathrm{start}}_{d}\,\tilde{E}^{D}_{d} \tag{34}$$

$$E^{\mathrm{SOC},D}_{d,s,|T|} = \gamma^{\mathrm{end}}_{d}\,\tilde{E}^{D}_{d} \tag{35}$$

- **Free end level.** Equation (32) applies for $t \geq 2$and Equation (34) applies, but Equation (35) is omitted. The reservoir may end the horizon at any permitted level. This is the appropriate choice for a drought study, and it will always report a lower cost than the cyclic case, because ending empty is free.

Because spill is unbounded, the cyclic condition is always feasible. Any inflow that cannot be stored or turbined is spilled, so no combination of inflow data and capacity can make Equation (32) infeasible under the cyclic closure.

10.3 Turbine and Pump Limits

Equation (36) limits turbine output to installed turbine capacity.

$$e^{D\,Gen}_{d,s,t} \leq P^{D\,Gen}_{d} \tag{36}$$

Equation (36) is the linear form. Where a plant has multiple units with a minimum stable output, it may be replaced by the binary or discrete unit formulations of Section 9, with $e^{D\,Gen}_{d,s,t}$substituted for $e_{n,j,s,t}$. The reservoir balance is unaffected by that choice.

Equation (37) limits pump consumption to installed pump capacity. For a conventional dam $P^{D\,Pump}_{d} = 0$, which forces $e^{D\,Pump}_{d,s,t} = 0$at every timestep.

$$e^{D\,Pump}_{d,s,t} \leq P^{D\,Pump}_{d} \tag{37}$$

10.4 Minimum Flow

Downstream flow requirements are a licensing condition at most sites and are usually the binding limit on how much operational flexibility a hydro plant can offer. Equation (38) requires total water leaving the reservoir to meet a minimum in every timestep. Both terms are expressed in stored energy potential, which is proportional to volume, so turbine release enters as $e^{D\,Gen}_{d,s,t}/\eta^{D}_{Gen}$rather than as delivered electricity.

$$\Phi_{d,t} \leq \frac{e_{d,s,t}^{D\,Gen}}{\eta_{Gen}^{D}} + \varsigma_{d,s,t} \tag{38}$$

$\Phi_{d,t}$ is the minimum release in kWh-equivalent, converted from the licensed volumetric requirement. It varies through the year and is indexed by t rather than being constant.

10.5 Assumptions

- **Constant head.** Expressing inventory in kWh implies a fixed conversion between stored water volume and energy, which holds only if hydraulic head is constant. The conversion coefficient should be set at the upper end of the operating head band, because a coefficient fixed at a lower head causes Equation (38) to be satisfied in energy terms while releasing less water than the license requires. State the drawdown range over which the approximation holds and the resulting error in volumetric terms.

- **Post-processing to volume.** Inventory, spill, and release are reported in kWh and converted to acre-feet or cfs after the solve, using the same coefficient. This conversion is exact only under the constant head assumption above. It is not equivalent to the hydrogen kWh to kg conversion, which uses a physical constant and is exact.

- **Single reservoir per unit.** Only the upper reservoir is tracked. For pumped storage the lower reservoir is assumed unconstrained, which is valid for an open loop plant on a river and requires revisiting for a closed loop plant.

- **No cascade.** Water released from one unit is not routed as inflow to another. Cascaded systems require a hydraulic arc set and a transit delay, neither of which is included.

- **Spillway capacity unbounded.** If spillway capacity is a binding physical limit, add $\varsigma_{d,s,t} \leq \bar{\varsigma}_{d}\ \Delta$. Note that doing so removes the guarantee in Section 10.2 that the cyclic condition is always feasible.

11Test Case

This section compares optimized microgrid designs for Igiugig, AK with legacy diesel generation available and with diesel generation removed. Both cases use the HydroFlex implementation configured to the baseline parameters of the manuscript. Results are

compared against the published values in Table 6 and Table 7 of a manuscript submitted for publication. Hydrokinetic turbine (HKT) capital cost is $4,000/kW in both cases. HKT deployment is limited to two 80 kW units per site. These tests use the mixed integer linear program (MILP) formulation of the model. The diesel generator uses the linear/binary model describe in Section 8.2

The Igiugig, AK microgrid with hydrokinetic generation is shown in Figure 1. This figure captures a microgrid system for a remove village and includes existing diesel generation and battery storage, with the option to adopt and expand solar PV generation, HKT, additional battery capacity, and a H_2 energy storage system consisting of an electrolyzer, H_2 compression and storage, and proton exchange membrane fuel cell (PEMFC). In this instance, the HydroFlex tool would be used to capture the interactions between the different modeled components to find the lowest cost solution for meeting the village electric demand. A key milestone in HydroFlex development is the comparison of the current model versus the prior DEROpt model to show that the current HydroFlex model replicates results produced using the proven model.

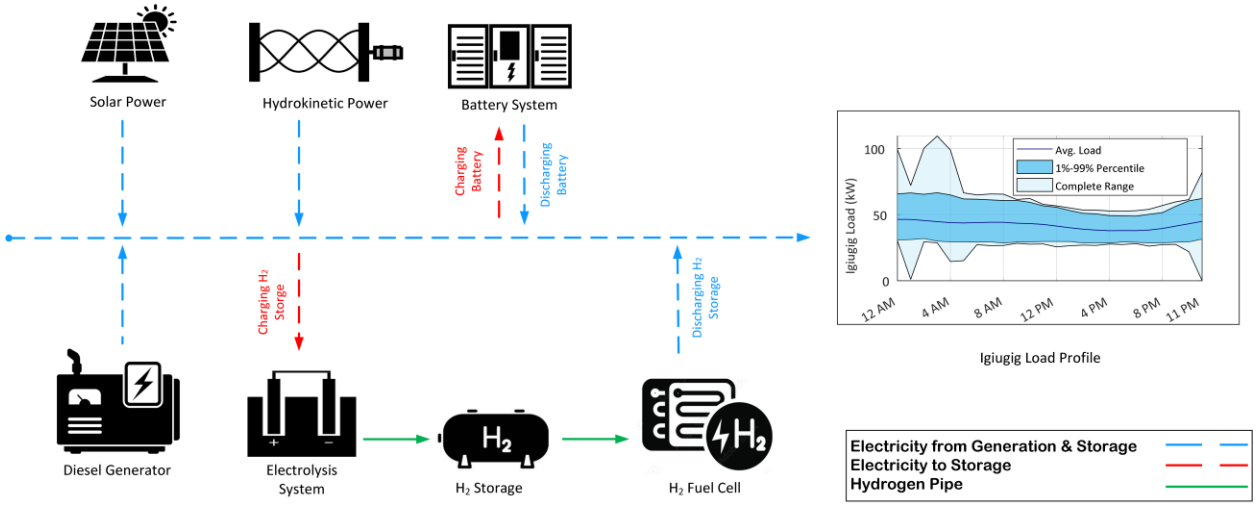


Figure 1: Schematic of the energy technologies and flows captured in an optimization model developed using the HydroFlex optimization code.

11.1 Performance Evaluation

MILP results are evaluated using cost and carbon metrics. Cost performance is quantified with the levelized cost of electricity (LCOE) in Equation (39). Carbon emissions are evaluated using carbon intensity (CI) defined in Equation (40). Since carbon is emitted on

from diesel engine operation, CI is calculated using a CI_{Diesel} emission factor 0.2705 kg CO_2 per kWh diesel fuel used.

$$LCOE = \frac{Cost\ Function}{\sum_t load_t} \tag{39}$$

$$CI = \frac{CI_{Diesel} \cdot \sum_t \frac{e_{Diesel,t}}{\eta_{Diesel}}}{\sum_t load_t} \tag{40}$$

11.2 Model Configuration

The optimization uses the parameter set of manuscript Tables 1 and 3. Capital costs are annualized at 6% over ten years, giving a capital recovery factor of 0.1359. The community load is 369,453 kWh/year with a mean of 42.2 kW and a peak of 109.5 kW. Legacy diesel capacity is three 50 kW generators and is treated as sunk capital, so only fuel cost and variable operations and maintenance cost are charged to diesel operation. The existing 250 kWh battery is carried in both cases.

Both cases were solved as the full mixed integer linear program on the 8,760 hour grid using Gurobi 13.0.1 with a relative MIP gap target of 0.5%. The diesel available case terminated at a gap of 0.49% in 230 seconds. The diesel free case terminated at a gap of 0.25% in 223 seconds. Because the diesel available case terminated on the gap criterion, its reported cost is an upper bound. Its LCOE lies between $0.4271/kWh and $0.4292/kWh.

A calibration check was run on the legacy diesel baseline of manuscript Table 5. The model returns an LCOE of $0.8248/kWh and a carbon intensity of 0.8198 kg CO2/kWh, against published values of $0.827/kWh and 0.822 kg CO2/kWh. These results differ by 0.3%.

Table 3: Adopted capacity, cost, and carbon intensity for the two cases

Quantity	Diesel available	Diesel free
Site 1 HKT units	2	2
Site 2 HKT units	0	2
Total HKT capacity (kW)	160	320
Single-axis tracking PV (kW)	60.7	140.8
New battery capacity (kWh)	4.9	50.1
PEM fuel cell (kW)	0	14.6
Electrolyzer (kW H2)	0	32.7
H2 storage (kg)	0	317.0
Annualized system cost ($/year)	158,568	279,722
LCOE ($/kWh)	0.429	0.757
Carbon intensity (kg CO2/kWh)	0.129	0
Renewable resource utilization	75.7%	58.2%

Table 4: Annual generation by source.

Source	Diesel available	Diesel free
HKT	66.1%	59.2%
Solar PV	18.4%	38.3%
Diesel	15.5%	0%
Fuel cell	0%	2.4%

11.3 Comparison of the Two Cases

This section discusses differences between the two cases tested in the HydroFlex model – "Diesel available" versus "Diesel free". The two designs are not equivalent. Removing diesel generation changes the technology selection, the system cost, and the utilization of the renewable resource.

Adopted HKT capacity doubles from 160 kW to 320 kW. With diesel available the optimization selects two units at Site 1 and does not select Site 2. With diesel removed it selects the maximum number of units at both sites.

Adopted PV capacity increases from 60.7 kW to 140.8 kW. Hydrogen technologies are not adopted when diesel is available and are adopted when diesel is removed. The adopted hydrogen system is a 32.7 kW H2 electrolyzer with a 14.6 kW fuel cell and 317 kg of hydrogen storage. New battery capacity increases from 4.9 kWh to 50.1 kWh, which is small relative to the existing 250 kWh system in both cases.

Annualized system cost increases by $121,154, which raises LCOE from $0.429/kWh to $0.757/kWh, an increase of 76%. Carbon intensity falls from 0.129 kg CO2/kWh to zero.

The 15.5% diesel energy share in the diesel available case is supplied at a fuel and variable operations and maintenance cost of $0.825/kWh. Replacing that share requires 160 kW of additional HKT capacity, 80.1 kW of additional PV capacity, and a hydrogen storage system. The cost of that replacement capacity exceeds the avoided diesel cost, which is the reason for the LCOE increase.

Resource utilization falls from 75.7% to 58.2%. In the diesel free case 289,595 kWh of available HKT and solar energy is not used, against 101,830 kWh in the diesel available case. Diesel generation acts as a firm supply that allows the renewable capacity to be sized close to what can be absorbed. Without diesel the renewable capacity must be sized for the lowest-resource periods, so a larger fraction of annual output has no load or storage available to receive it.

11.4 Comparison Against DERopt

Technology selection matches the manuscript in both cases. HKT unit counts are identical at both sites. Hydrogen technologies are adopted in the same case and declined in the same case.

LCOE agrees to within 0.6%. The diesel available value is 0.4% below the published value and the diesel free value is 0.6% below. Carbon intensity in the diesel available case is 2.3% above the published value.

Adopted capacities agree to within 18%. PV capacity is 8% low in the diesel available case and 2% low in the diesel free case. Fuel cell capacity is 9% low, electrolyzer capacity is 18% low, and hydrogen storage capacity is 15% high. The direction of the hydrogen differences is consistent across the three components. The model selects a smaller conversion capacity operating for longer periods together with a larger storage vessel, while the manuscript selects the reverse. Total annualized hydrogen system cost differs by less than the individual component differences suggest, because the storage vessel is inexpensive per unit energy at $25/kWh H2 relative to the conversion equipment.

The relative effect of removing diesel is reproduced more closely than any absolute value. The model gives a diesel free to diesel available LCOE ratio of 1.764. The manuscript values give a ratio of 1.768, a difference of 0.2%.

New battery capacity shows the largest proportional disagreement. The model adopts 50.1 kWh against a published 94 kWh in the diesel free case, and 4.9 kWh against a published zero in the diesel available case. Both values are small relative to the existing 250 kWh system, so the effect on system cost is limited. At $830/kWh the 44 kWh difference is $36,500 of capital, or $4,960/year, which is 1.8% of diesel free system cost.

Table 5: Model results against the results from the original DERopt model.

Quantity	Diesel available, HydroFlex	Diesel available, DERopt	Diesel free, HydroFlex	Diesel free, DERopt

Site 1 HKT units	2	2	2	2
Site 2 HKT units	0	0	2	2
Tracking PV (kW)	60.7	66	140.8	144
New battery (kWh)	4.9	0	50.1	94
PEM fuel cell (kW)	0	0	14.6	16
Electrolyzer (kW H2)	0	0	32.7	40
H2 storage (kg)	0	0	317.0	276
LCOE ($/kWh)	0.429	0.431	0.757	0.762
Carbon intensity (kg CO2/kWh)	0.129	0.126	0	0

11.5 Limitations

The diesel available case terminated on the 0.5% MIP gap criterion rather than proven optimality. Its true optimum lies between $0.4271/kWh and $0.4292/kWh. Adopted PV capacity in that case may shift with a tighter gap, so the 60.7 kW value should be confirmed at a gap of 0.1% before it is quoted as a design value.

Three variable operations and maintenance terms in the manuscript have no equivalent parameter in the current HydroFlex technology blocks and were set to zero. These are battery throughput cost of $0.001/kWh, PV variable cost of $0.001/kWh, and hydrogen storage charge and discharge cost of $0.01/kWh and $0.001/kWh. The combined effect on LCOE is below $0.005/kWh at the adopted capacities. This omission accounts for part of the residual LCOE difference against the manuscript. These changes are due to actual operations and maintenance for batteries, solar PV, and hydrogen tanks, which occur each year regardless of usage.

12References

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